/// CURRICULUM VITÆ

RAPHAËL LEMAITRE.

Backend engineer with 26 years on the JVM, including about ten years of functional Scala. I'm a Senior Backend Engineer at Nabla, building the real-time dictation backend in Kotlin. I have been a CTO and went back to coding by choice. I care about type safety, simple systems and teams that ship. On the side, I speak at Scala conferences and maintain open-source Scala libraries.

PARIS AREA, FR
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01 EXPERIENCE

2025.11 → NOW

SENIOR BACKEND ENGINEER

Nabla — Dictation squad KOTLIN · POSTGRESQL · GOOGLE CLOUD · STT MODELS (WHISPER, NEMOTRON, SPEECHMATICS)

- Refactored the transcription pipeline to separate dictation from ambient transcription, by usage type.
- Designed and built a layered, rule-based formatter with entity detection, in English and French, replacing the regex-based one.
- Built transcription A/B testing: each transcription runs on the production pipeline and a candidate one, and an LLM judge compares the results without exposing PHI.
- Wrote an internal API client so the ambient product uses the Dictation API without depending on contracts that may change.
- Handled changes of where the dictated text lands without restarting the transcription pipeline.
- Design and code reviews, coding and system design interviews, AI-assisted coding workflows.

2022.03 → 2025.10

SENIOR STAFF ENGINEER

Ledger SCALA · CATS EFFECT · POSTGRESQL

- Architected ATLAS, Ledger's blockchain indexing platform for fast transaction and balance access.
- Built high-availability Scala backend services for Ledger's core products.
- Ran PostgreSQL at scale and a fleet of blockchain nodes.
- Analyzed and optimized cryptocurrency protocols to improve data indexing and query speeds.
- Benchmarked services to find and fix bottlenecks, with CI/CD and monitoring behind every deployment.

2014.06 → 2022.03

CTO, PREVIOUSLY LEAD DEVELOPER

Adventori, now part of
the COOL company GROOVY · GRAILS · JAVASCRIPT · POSTGRESQL · CASSANDRA · APACHE SPARK

- Served 20 billion ads a year on 15 servers at 99.5% availability, and ingested 400 GB of data a day on 20 servers.
- Managed the Tech Lab, a team of 9 developers, DevOps, data engineers and integrators.
- Co-designed the product roadmap with the CEO and PM, and aligned the technical roadmap with company objectives.
- Mentored the team through peer reviews, katas and coding dojos, and promoted craftsmanship and DevOps practices.
- Built an inclusive and selective hiring process for engineering and product.
- As Lead Developer, designed and built the ad server, data warehouse, reporting system and customer data integration service.
- Introduced Cassandra for high-volume data and Apache Spark for data ingestion and reporting.
- Set up advanced CI/CD workflows.

2010.10 → 2014.06

R&D SENIOR DEVELOPER

Systar, now part of Axway JAVA · FLEX (ACTIONSCRIPT)

— Designed and built a monitoring application on a custom distributed, bi-temporal NoSQL database.
— Built calculated fields management and optimized aggregation computations (OLAP cube).
— Scrum Master for a team of 4 developers in Paris and Lyon.
— Built a JDBC driver for the database and a quantified-self monitoring application during hacking weeks.

2007.05 → 2010.08

R&D ENGINEER

Xcalia, now part of Progress Software JAVA · SQL · XQUERY

— Developed a datastore-agnostic persistence and query engine over RDBMS and web services.
— Implemented compensation algorithms (joins, sort, union) and datastore-specific query generation.
— Built dynamic service orchestration, independent of the service invocation technology.

2006.07 → 2007.05

SOFTWARE ARCHITECT

Capgemini JAVA · JBOSS · JASPERREPORTS · ORACLE

— Designed a portal and directory architecture for an AP-HP call for bids.
— Led the migration of the DGI's FICOBA application from Oracle Application Server to JBoss and from Oracle Reports to JasperReports.

2000.01 → 2006.06

TEAM LEADER

E-Deal, now part of Efficy JAVA · J2EE · J2ME · POSTGRESQL · ORACLE · DB2

— Managed a team of 4 R&D developers.
— Built the first J2EE version of the e-DEAL CRM, with its MVC and persistence frameworks, Sales Force and Admin modules.
— Added internationalization, calendar and contact sync with Microsoft Exchange and Lotus Domino, and a J2ME application.
— Rewrote the presentation framework (MVC2).

02 SKILLS

LANGUAGES
Scala, Kotlin, Java, Groovy, SQL, TypeScript, JavaScript, HTML, CSS, Bash

AI
Claude Code, Cursor, MCP servers, Prompt engineering, Anthropic, OpenAI, Gemini, STT (Whisper, Nemotron, Speechmatics, Soniox)

DATA
PostgreSQL, SQLite, Apache Cassandra, Apache Spark, Elasticsearch

OPS & TOOLS
Kubernetes, Docker, Gatling, Git, GitHub, GitLab, CI/CD, Jenkins, Linux, OpenTelemetry, Grafana, Kibana, Datadog, IntelliJ IDEA, VS Code, Metals, Zed

03 TALKS

×3 · 2024–2025
From Zero to Production Faster Than Your Average Meeting With Pillars
FUNCTIONAL SCALA · SCALAR · SCALAIO

×5 · 2023–2024
Armored Type Safety With Iron
SCALA MATTERS · SCALAIO · SCALAR · SCALAMATSURI · FUNCTIONAL SCALA

×1 · 2024
Fearless Coding: Scala 3 and Iron, the New Heroes of Reliability in Functional Programming
SUNNYTECH

×2 · 2024
Hacking ADHD: Strategies for the Modern Developer
DEVOXX FRANCE · SUNNYTECH

04 EDUCATION

1997 → 2001
MASTER IN ENGINEERING, COMPUTER SCIENCE ENSIIE

05 CERTIFICATIONS

2015 — **INTRODUCTION TO BIG DATA WITH APACHE SPARK** edX
2013 — **PRINCIPLES OF REACTIVE PROGRAMMING** Coursera
2012 — **FUNCTIONAL PROGRAMMING WITH SCALA** Coursera